

Probability: With & Without Replacement

Bag: 3 Red, 4 Blue, 5 Green = 12 balls

1. Draw 2 balls without replacement. What is the probability that the first ball is Red and the second ball is Blue?

Solution

$$\frac{{}^3C_1}{{}^{12}C_1} \times \frac{{}^4C_1}{{}^{11}C_1} = \frac{1}{11}$$

OR

$$\frac{3}{12} \times \frac{4}{11} = \frac{1}{11}$$

2. Draw 2 balls with replacement. What is the probability that the first ball is Red and the second ball is Blue?

Solution

$$\frac{{}^3C_1}{{}^{12}C_1} \times \frac{{}^4C_1}{{}^{12}C_1} = \frac{1}{12}$$

OR

$$\frac{3}{12} \times \frac{4}{12} = \frac{1}{12}$$

3. Draw 2 balls without replacement. What is the probability of getting exactly 1 Red and 1 Blue?

Solution

$$\frac{{}^3C_1 \times {}^4C_1}{{}^{12}C_2} = \frac{2}{11}$$

OR

$$\left(\frac{3}{12} \times \frac{4}{11}\right) + \left(\frac{4}{12} \times \frac{3}{11}\right) = \frac{2}{11}$$

$$\frac{2!}{1! \times 1!} \times \left(\frac{3}{12} \times \frac{4}{11}\right) = \frac{2}{11}$$

4. Draw 2 balls with replacement. What is the probability of getting exactly 1 Red and 1 Blue?

Solution

$$\frac{2!}{1! \times 1!} \times \frac{{}^3C_1}{{}^{12}C_1} \times \frac{{}^4C_1}{{}^{12}C_1} = \frac{1}{6}$$

OR

$$\left(\frac{3}{12} \times \frac{4}{12}\right) + \left(\frac{3}{12} \times \frac{4}{12}\right) = \frac{1}{6}$$

$$\frac{2!}{1! \times 1!} \times \left(\frac{3}{12} \times \frac{4}{12}\right) = \frac{1}{6}$$

5. Draw 3 balls without replacement. What is the probability of getting exactly 2 Red and 1 Green?

Solution

$$\frac{{}^3C_2 \times {}^5C_1}{{}^{12}C_3} = \frac{3}{44}$$

OR

$$\left(\frac{3}{12} \times \frac{2}{11} \times \frac{5}{10}\right) + \left(\frac{5}{12} \times \frac{3}{11} \times \frac{2}{10}\right) + \left(\frac{3}{12} \times \frac{5}{11} \times \frac{2}{10}\right) = \frac{3}{44}$$

$$\frac{3!}{2! \times 1!} \times \left(\frac{3}{12} \times \frac{2}{11} \times \frac{5}{10}\right) = \frac{3}{44}$$

6. Draw 3 balls with replacement. What is the probability of getting exactly 2 Red and 1 Green?

Solution

$$\frac{{}^3C_1}{{}^{12}C_1} \times \frac{{}^3C_1}{{}^{12}C_1} \times \frac{{}^5C_1}{{}^{12}C_1} + \frac{{}^3C_1}{{}^{12}C_1} \times \frac{{}^5C_1}{{}^{12}C_1} \times \frac{{}^3C_1}{{}^{12}C_1} + \frac{{}^5C_1}{{}^{12}C_1} \times \frac{{}^3C_1}{{}^{12}C_1} \times \frac{{}^3C_1}{{}^{12}C_1} = \frac{5}{64}$$

$$\frac{3!}{2! \times 1!} \times \left(\frac{{}^3C_1}{{}^{12}C_1}\right)^2 \times \frac{{}^5C_1}{{}^{12}C_1} = \frac{5}{64}$$

OR

$$\frac{3!}{2! \times 1!} \times \left(\frac{3}{12} \times \frac{3}{12} \times \frac{5}{12}\right) = \frac{5}{64}$$

7. Draw 4 balls without replacement. What is the probability of getting exactly 2 Blue and 2 Green?

Solution

$$\frac{{}^4C_2 \times {}^5C_2}{{}^{12}C_4} = \frac{4}{33}$$

OR

$$\frac{4!}{2! \times 2!} \times \left(\frac{4}{12} \times \frac{3}{11} \times \frac{5}{10} \times \frac{4}{9}\right) = \frac{4}{33}$$

8. Draw 4 balls with replacement. What is the probability of getting exactly 2 Blue and 2 Green?

Solution

$$\frac{4!}{2! \times 2!} \times \left(\frac{4C1}{12C1}\right)^2 \times \left(\frac{5C1}{12C1}\right)^2 = \frac{25}{216}$$

OR

$$\frac{4!}{2! \times 2!} \times \left(\frac{4}{12} \times \frac{4}{12} \times \frac{5}{12} \times \frac{5}{12}\right) = \frac{25}{216}$$

9. Draw 3 balls without replacement. What is the probability of getting exactly 1 Red, 1 Blue and 1 Green?

Solution

$$\frac{3C1 \times 4C1 \times 5C1}{12C3} = \frac{3}{11}$$

OR

$$\frac{3!}{1! \times 1! \times 1!} \times \left(\frac{3}{12} \times \frac{4}{11} \times \frac{5}{10}\right) = \frac{3}{11}$$

10. Draw 3 balls with replacement. What is the probability that all 3 balls are Blue?

Solution

$$\frac{3!}{3!} \times \left(\frac{4C1}{12C1}\right)^3 = \frac{1}{27}$$

OR

$$\frac{3!}{3!} \times \left(\frac{4}{12}\right)^3 = \frac{1}{27}$$

11. Draw 3 balls without replacement. What is the probability of getting at least 1 Red?

Solution

$$1 - \frac{9C3}{12C3} = \frac{34}{55}$$

OR

$$1 - \frac{3!}{3!} \times \left(\frac{9}{12} \times \frac{8}{11} \times \frac{7}{10}\right) = \frac{34}{55}$$

12. Draw 3 balls without replacement. What is the probability of getting at most 1 Red?

Solution

$$\frac{{}^9C_3}{{}^{12}C_3} + \frac{{}^3C_1 \times {}^9C_2}{{}^{12}C_3} = \frac{48}{55}$$

OR

$$\left(\frac{3!}{3!} \times \frac{9}{12} \times \frac{8}{11} \times \frac{7}{10} + \frac{3!}{1! \times 2!} \times \frac{3}{12} \times \frac{9}{11} \times \frac{8}{10}\right) = \frac{48}{55}$$

13. Draw 2 balls without replacement. What is the probability that both balls are Green?

Solution

$$\frac{{}^5C_2}{{}^{12}C_2} = \frac{5}{33}$$

OR

$$\frac{2!}{2!} \times \frac{5}{12} \times \frac{4}{11} = \frac{5}{33}$$

14. Draw 4 balls without replacement. What is the probability of getting exactly 2 Red and 2 Blue?

Solution

$$\frac{{}^3C_2 \times {}^4C_2}{{}^{12}C_4} = \frac{2}{55}$$

OR

$$\frac{4!}{2! \times 2!} \times \left(\frac{3}{12} \times \frac{2}{11} \times \frac{4}{10} \times \frac{3}{9}\right) = \frac{2}{55}$$